Chapter 3 — Circular and Rotational Motion

Complete Notes · PECTAA Physics 11 · Board-style longs + conceptual shorts

Topic 1 — Angular displacement, velocity and acceleration

Q.1 Define the radian and derive the relations between linear and angular quantities.

One radian is the angle subtended at the centre of a circle by an arc equal in length to the radius: $\theta = s/r$. Since the circumference is $2\pi r$, a full circle is 2π rad = 360°, so 1 rad ≈ 57.3°. Angular displacement θ is measured in radians, angular velocity $\omega = \theta/t$ in rad s^{-1}, and angular acceleration $\alpha = \Delta\omega/t$ in rad s^{-2}.

For a body moving in a circle of radius r, the linear and angular quantities are related by $s = r\theta$, $v = r\omega$ and $a_t = r\alpha$. The equations of angular motion for uniform α are $\omega_f = \omega_i + \alpha t$, $\theta = \frac{1}{2}(\omega_i + \omega_f)t$ and $\omega_f^2 = \omega_i^2 + 2\alpha\theta$, exactly analogous to the linear equations.

Important Conceptual Short Questions — Topic 1

1. Why is the radian a dimensionless unit?

A radian is the ratio of arc length to radius, both measured in metres. The ratio [L]/[L] has no dimensions, so the radian is dimensionless although it is written to show that an angle is being used.

2. How is linear speed related to angular speed?

In one revolution a point travels a distance $2\pi r$ while the angle swept is 2π. Therefore $v = r\omega$: points farther from the axis have greater linear speed for the same angular speed.

3. Why does a body in uniform circular motion have acceleration?

Although the speed is constant, the direction of velocity changes continuously. A change in velocity means acceleration, directed toward the centre of the circle.

Topic 2 — Centripetal force and circular motion

Q.2 Derive centripetal acceleration and centripetal force and give examples of their sources.

A body moving with speed v in a circle of radius r has an acceleration toward the centre of magnitude $a_c = v^2/r = r\omega^2$. By Newton's second law the required inward force is $F_c = mv^2/r = mr\omega^2$. This force is not a new kind of force; it is supplied by an existing force such as tension, friction, gravity or the normal reaction.

Examples include the tension in a string whirling a stone, friction between tyres and road for a car turning a curve, gravity keeping a satellite in orbit, and the normal force on a passenger in a rotating space station. If the centripetal force

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disappears, the body moves off along the tangent, not radially outward.

Important Conceptual Short Questions — Topic 2

1. Why does centripetal force do no work in uniform circular motion?

The force is always perpendicular to the instantaneous displacement, which is along the tangent. Since W = Fd cos 90° = 0, the force changes only the direction of velocity, not the speed or kinetic energy.

2. Why does a stone fly off tangentially when the string breaks?

At the instant the string breaks the velocity is along the tangent. With no centripetal force there is no inward acceleration, so by Newton's first law the stone continues along the tangent.

3. How does friction provide centripetal force for a turning car?

The tyres tend to slide outward relative to the road. Static friction acts inward toward the centre of the curve and supplies the required mv^2/r.

Topic 3 — Artificial satellites and artificial gravity

Q.3 Derive the orbital speed and period of an artificial satellite and explain artificial gravity.

For a satellite of mass m orbiting the Earth of mass M at radius r, gravity provides the centripetal force: $GMm/r^2 = mv^2/r$. Hence $v = \sqrt{(GM/r)}$ and $T = 2\pi r/v = 2\pi\sqrt{(r^3/GM)}$. The mass of the satellite cancels, so orbital speed depends only on the mass of the Earth and the orbital radius. Close to the Earth's surface $v \approx 7.9$ km s^{-1} and $T \approx 84$ min.

In a rotating space station the normal force from the wall provides the centripetal force and feels like weight. The frequency needed to simulate gravity is $f = (1/2\pi)\sqrt{(g/R)}$, where R is the radius of the station. Astronauts in orbit feel weightless because they and the satellite are falling freely with the same acceleration.

Textbook check (p. 57): Example 3.4 prints the orbital speed as 1.025 km s^{-1}, which is the value before taking the square root. The correct speed is about 1.01 km s^{-1} and the period about 27.7 days.

Important Conceptual Short Questions — Topic 3

1. Why does the mass of a satellite not affect its orbital speed?

The satellite mass appears on both sides of $GMm/r^2 = mv^2/r$ and cancels. The orbital speed therefore depends only on the central mass and the radius.

2. Why do astronauts in orbit feel weightless?

The astronaut and the satellite are both falling freely toward the Earth with the same acceleration. There is no normal force between them, so the astronaut feels no weight.

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3. How can a rotating space station produce artificial gravity?

The wall of the station pushes the astronaut inward with a normal force that provides centripetal acceleration. The astronaut feels this push as weight, just as the floor pushes us upward on Earth.

Topic 4 — Moment of inertia and torque

Q.4 Explain moment of inertia and derive the relation between torque and angular acceleration.

Moment of inertia is the rotational analogue of mass: $I = \Sigma mr^2$, where r is the perpendicular distance of each mass element from the axis. It depends not only on the total mass but also on how the mass is distributed relative to the axis. Torque is the turning effect of a force: $\tau = rF \sin \theta$. For a rigid body, applying $F = ma$ to each mass element and summing gives $\tau = I\alpha$, the rotational form of Newton's second law.

Linear quantity	Rotational analogue
Mass m	Moment of inertia $I = \Sigma mr^2$
Force F	Torque $\tau = rF \sin \theta$
Acceleration a	Angular acceleration α
Newton's second law F = ma	$\tau = I\alpha$

Important Conceptual Short Questions — Topic 4

1. Why does moment of inertia depend on the axis of rotation?

Moment of inertia is Σmr^2, and r is measured from the chosen axis. Changing the axis changes the distances of the mass elements, so the same body can have different moments of inertia.

2. How does torque differ from force?

Force produces linear acceleration, while torque produces angular acceleration. Torque depends on the force, the distance from the axis and the angle, $\tau = rF \sin \theta$.

3. Why does a flywheel store rotational energy?

A flywheel has a large moment of inertia. Work done to spin it is stored as rotational kinetic energy $\frac{1}{2}I\omega^2$, which can be released when the wheel slows down.

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Topic 5 — Angular momentum and its conservation

Q.5 State the law of conservation of angular momentum and explain its applications.

Angular momentum of a rotating body is L = Iω. If no external torque acts on a system, its total angular momentum remains constant. This is the law of conservation of angular momentum. When a skater pulls in the arms, I decreases, so ω increases to keep L constant. Divers and gymnasts use the same idea to spin faster in a tucked position.

Gyroscopes maintain their direction because their angular momentum resists change, which is used in navigation. Flywheels store energy in rotational motion, and centrifuges use large centripetal forces to separate materials of different densities.

Important Conceptual Short Questions — Topic 5

1. Why does a skater spin faster when the arms are pulled in?

Pulling in the arms reduces the moment of inertia. With no external torque, angular momentum Iω is conserved, so the angular velocity must increase.

2. How does a gyroscope maintain its direction?

A spinning gyroscope has a large angular momentum along its axis. Changing that direction requires an external torque, so in the absence of torque the axis remains fixed in space.

3. Why is angular momentum conserved in the absence of external torque?

Torque is the rate of change of angular momentum. If the external torque is zero, the angular momentum cannot change, so it remains constant.

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